

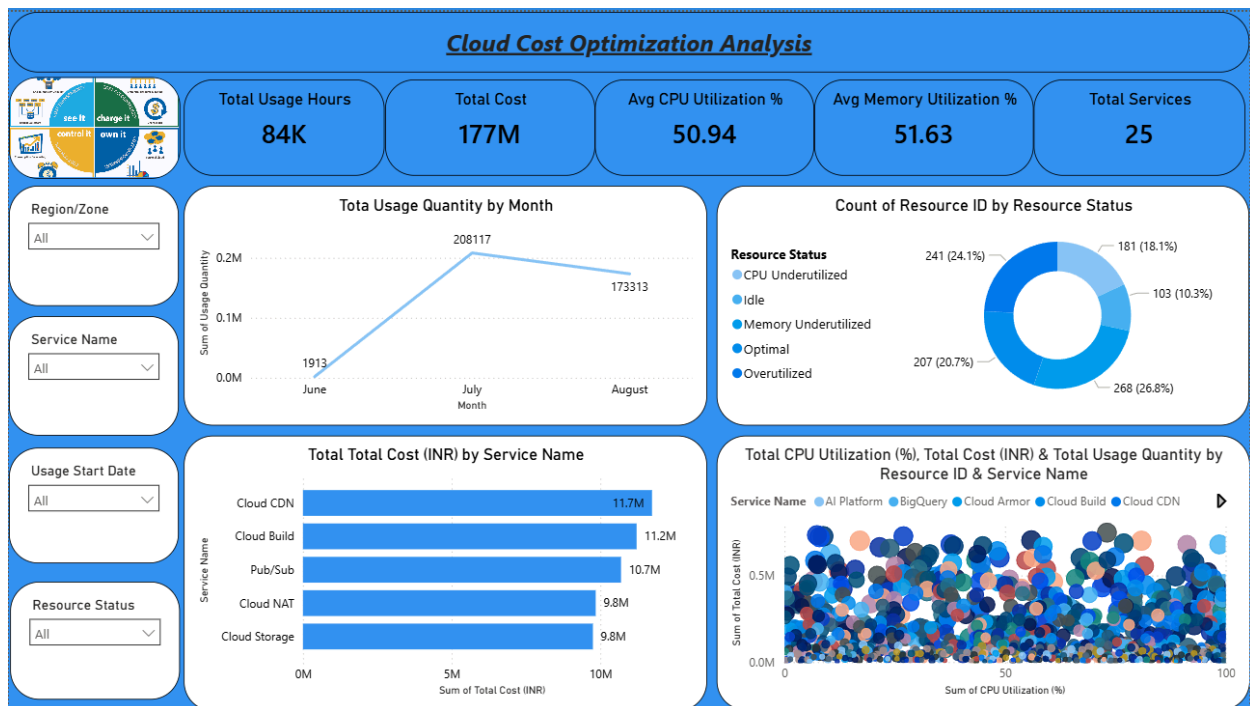
Cloud Cost Optimization Dashboard Report

1. Overview

The **Cloud Cost Optimization Dashboard** provides a comprehensive view of cloud resource usage, cost distribution, and efficiency. It helps identify cost drivers, monitor utilization trends, and detect underutilized resources to support optimization decisions.

The dashboard is designed to answer key business questions:

- How is cloud usage changing over time?
- Which services contribute the most to cost?
- Are there idle or underutilized resources?
- Where can cost optimization be applied?



2. Key Performance Indicators (KPIs)

The top section of the dashboard highlights important summary metrics:

Total Usage Hours	Total Cost	Avg CPU Utilization %	Avg Memory Utilization %	Total Services
84K	177M	50.94	51.63	25

- **Total Usage Hours: 84K**
Represents the total time resources were actively used.
- **Total Cost: 177M INR**
Overall cloud expenditure across all services.
- **Average CPU Utilization: 50.94%**
Indicates moderate CPU usage across resources.
- **Average Memory Utilization: 51.63%**
Shows balanced memory consumption.
- **Total Services: 25**
Number of distinct cloud services in use.

These KPIs provide a quick snapshot of overall cloud performance and cost efficiency.

3. Dashboard Visualizations

3.1 Usage Trend Analysis

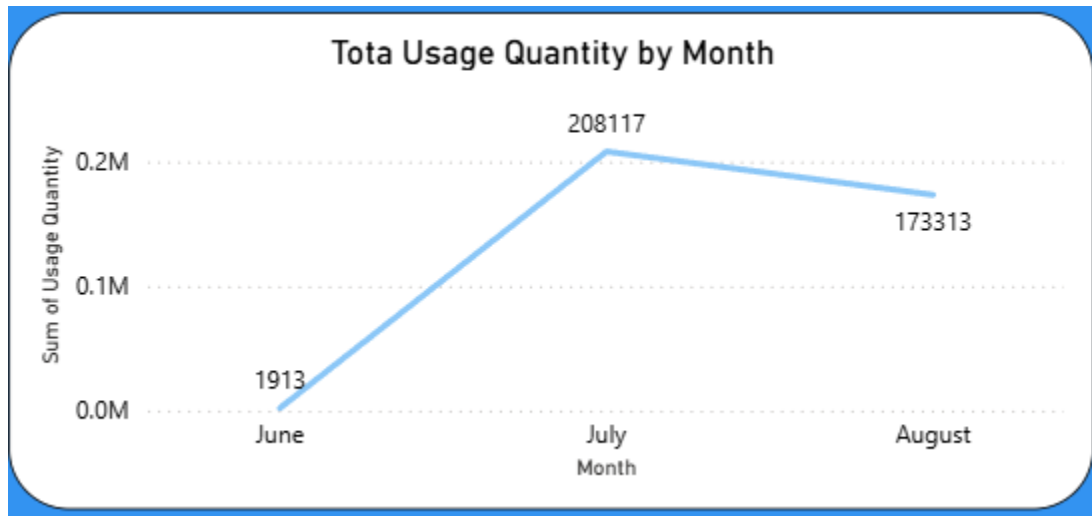


Chart: Total Usage Quantity by Month (Line Chart)

- Shows usage growth across months (June, July, August).
- Significant increase observed from June to July, followed by slight decline in August.

Insight:

- Indicates peak usage period in July.
- Helps correlate usage spikes with cost increases.

3.2 Resource Status Distribution

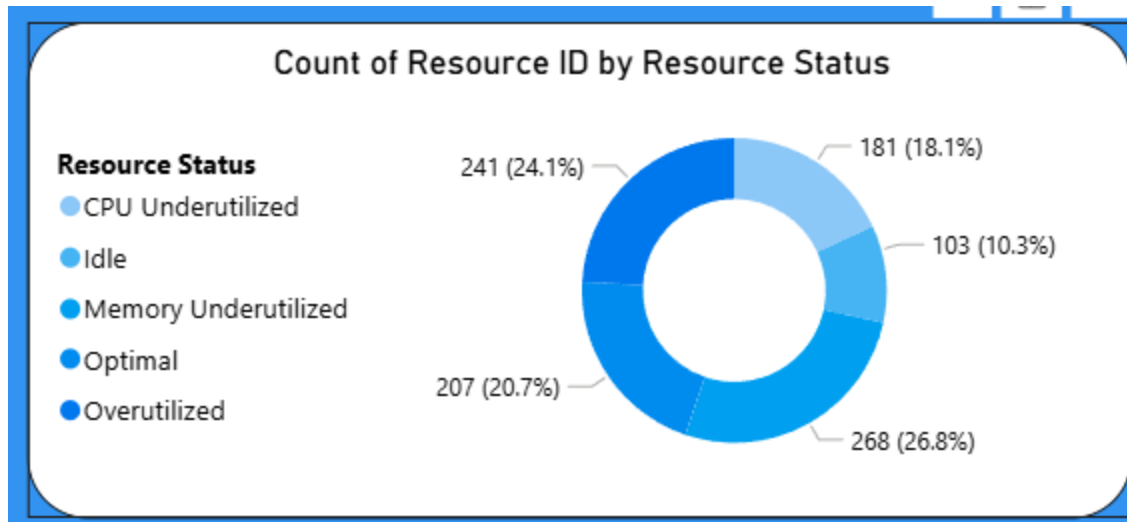


Chart: Resource Status Distribution (Donut Chart)

Categories include:

- CPU Underutilized
- Memory Underutilized
- Idle
- Optimal
- Overutilized

Insight:

- A considerable portion of resources falls under underutilized and idle categories.
- Highlights inefficiencies in resource allocation.

3.3 Cost by Service

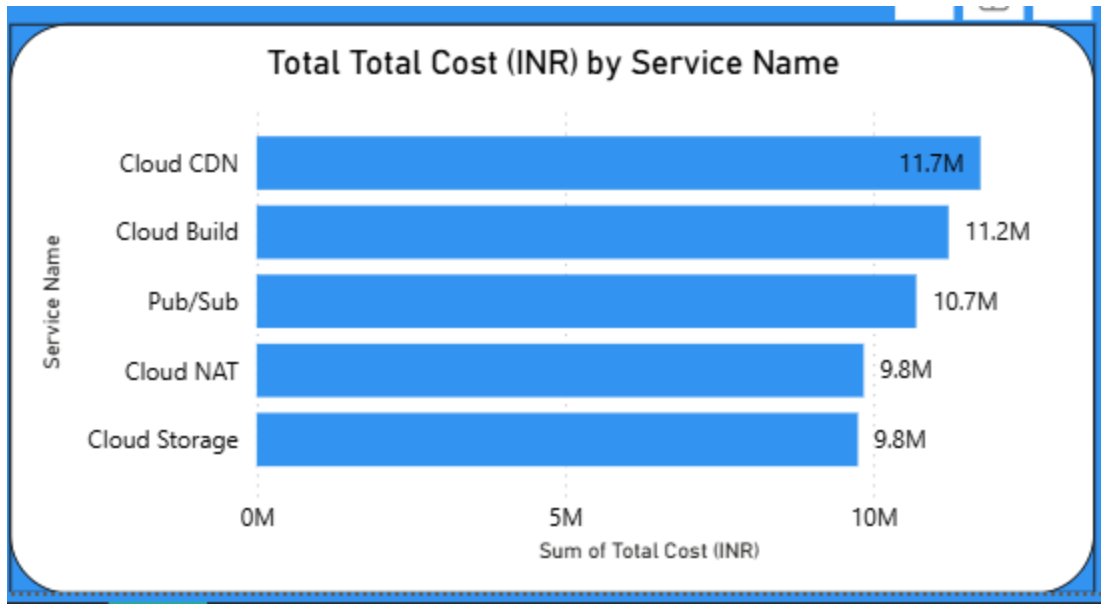


Chart: Total Cost (INR) by Service Name (Bar Chart)

Top cost-driving services:

- Cloud CDN (~11.7M)
- Cloud Build (~11.2M)
- Pub/Sub (~10.7M)
- Cloud NAT (~9.8M)
- Cloud Storage (~9.8M)

Insight:

- Identifies major contributors to cloud spending.
- Helps prioritize cost optimization efforts.

3.4 Resource Utilization vs Cost

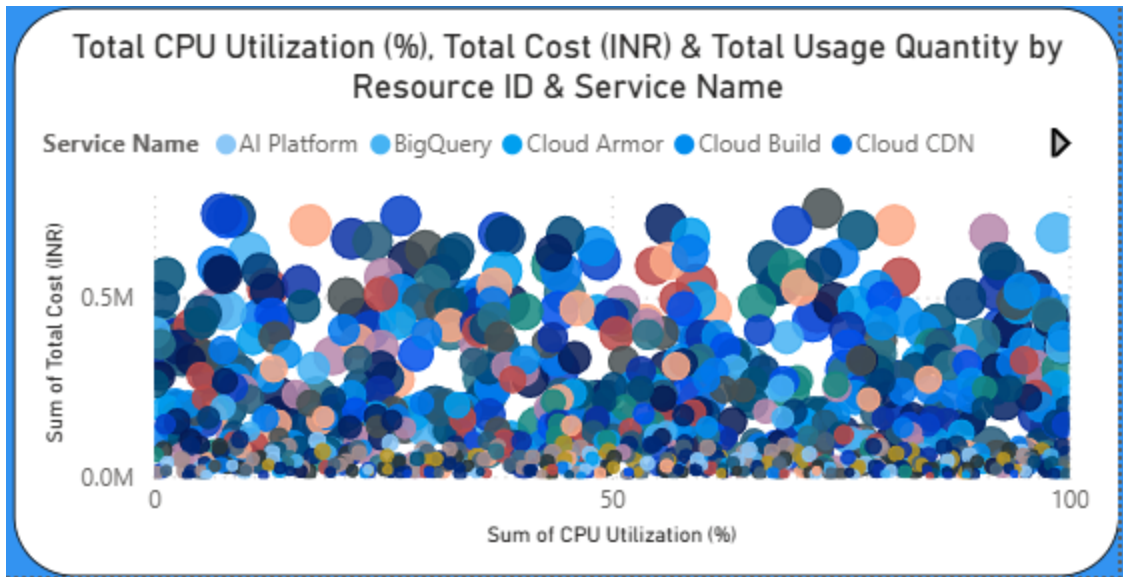


Chart: Scatter Plot (CPU Utilization vs Total Cost)

- Each point represents a resource.
- X-axis: CPU Utilization
- Y-axis: Total Cost
- Size: Usage Quantity
- Color: Service Name

Insight:

- Reveals resources with:
 - High cost but low utilization → **inefficient**
 - High cost and high utilization → **justified usage**
- Enables identification of over-provisioned resources.

4. Slicers (Filters)



The dashboard includes interactive slicers for dynamic analysis:

- **Region/Zone** → Analyze cost and usage geographically
- **Service Name** → Focus on specific services
- **Usage Start Date** → Time-based filtering
- **Resource Status** → Filter by Idle, Optimal, Underutilized

Purpose:

- Enhances interactivity
 - Allows drill-down analysis
 - Supports targeted decision-making
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5. Key Insights

- Cloud usage peaked in July, leading to higher costs.
 - Several services contribute significantly to total cost.
 - A notable percentage of resources are underutilized or idle.
 - Some high-cost resources show low utilization, indicating inefficiency.
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6. Conclusion

The Cloud Cost Optimization Dashboard effectively provides visibility into usage patterns, cost distribution, and resource efficiency.

It enables:

- Identification of high-cost services
- Detection of idle and underutilized resources
- Analysis of usage trends over time
- Data-driven cost optimization decisions

Overall, the dashboard serves as a powerful tool for improving cloud cost efficiency and resource management.